

Socialnetwork Diffusion with LLMs

An LLM agent-based model over the BCDJ microfinance networks

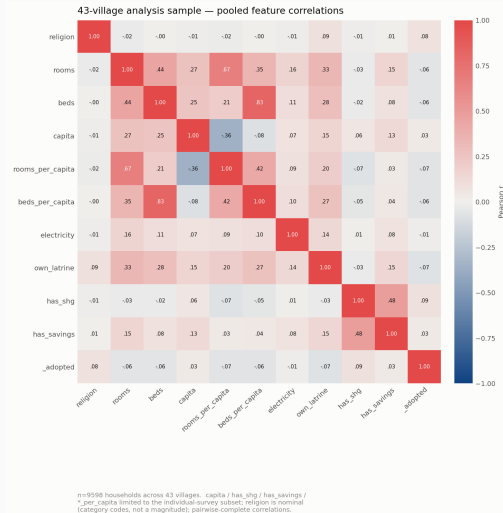
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Data interpretation

Modelling choices

- Agents are **households**; the individual survey covers a median 46.8% of a village.
- Covariates kept as **narrative grounding**, not as predictors: wealth proxies, financial exposure (SHG, savings), in/out-group standing (subcaste, religion).
- leader is **implied** in adoption, **explicit** in transmission – it is what the seeding routine conditions on; adopted likewise becomes conditional there.
- Edge type dropped**: nothing records who told whom, so there is no ground truth for which of the 12 relations carried influence. We run on the **union network**.
- Also dropped: redundant, high-missingness, redacted, near-constant and bare-identifier fields.



Two decision-theoretic games

Transmission $G^T = \langle A, S, P_i, U_i \rangle$

Will this household inform the neighbour?

- $A = \{\text{YES}, \text{NO}\}$
- $S = \{\text{THEY JOIN}, \text{THEY DON'T JOIN}\}$
- U_i : subjective risk/reward; P_i : beliefs over S

Played by every informed household, each turn, for each not-yet-informed / not-yet-adopted neighbour $j \in N$.

Assumption: transmission is **not a neutral act**. Consequences resemble a coordination game — *if I joined and told them, and they join too, I gain.*

Wider scope: an action space with intent — ENDORSE / INFORM / OPPOSE / IGNORE.

Adoption $G^A = \langle A, S, P_i, U_i \rangle$

Will this household adopt microfinance?

- $A = \{\text{YES}, \text{NO}\}$
- $S = \{\text{BENEFICIAL}, \text{LIMITED}, \text{HARMFUL}\}$
- U_i : subjective risk/reward; P_i : beliefs over S

Played once, whenever a household becomes informed.

S is the time-agnostic coarsening of what a loan does to a household's finances: positive (house improvement), neutral (balance transfer), negative (misuse, debt spiral).

BCDJ factorise diffusion into the same two conditionally independent decisions — a 7-parameter logit and a 2-parameter Bernoulli. We substitute one factor at a time.

Agent architecture: a structured wrapper, not off-the-shelf agency

1. The loop I would build — OODA

- **Observe** — neighbour states, what was said, village context
- **Orient** — household profile, priors, in/out-group standing
- **Decide** — the game above, over an explicit utility structure
- **Act** — adopt, or tell a specific neighbour

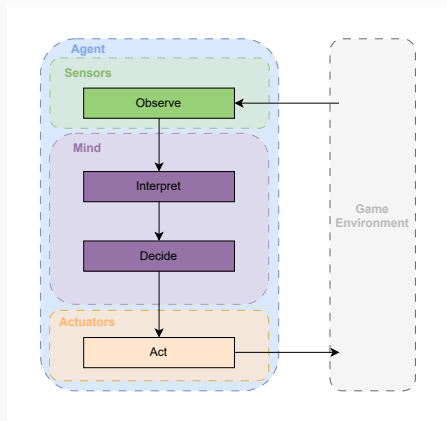
Models cognitive *and* physical features and carries environmental uncertainty explicitly. Strongly structured architectures (old-school Wooldridge) **wrapping** LLMs, rather than assuming inherent agency.

2. The loop actually implemented

- **Receive** — one transmission is delivered
- **Decide** — one LLM call, one decision
- **Advance** — state update, next round

Loosely OODA-shaped, but sequential and stateless. Full OODA was cut for time and complexity, not because it is the wrong model.

Simplifying assumptions taken from the original authors to save runtime: agents are informed once, decide once, and process one transmission at a time.



What varies: three axes

1. Information set

What the agent is told.

- Own profile (A)
- Neighbour's profile (B)
- Endorsement (C) / leader (L)

Each at three levels of richness:
absent, listed facts, or an LLM-written
narrative fixed per household id.

2. Instruction design

How the agent is asked to reason.

- Plain (D0)
- Model of appropriateness (D1)
- Decision-theoretic (D2)

Drawn from cognitive theory and
decision theory — not prompt
engineering folklore.

3. LLM choice

Who is answering.

- gpt-4.1-nano, gpt-5-nano
- gpt-5.4-nano (primary)
- gpt-5.6-luna
- grok-4.20, grok-4.1-fast

Non-reasoning models, so hidden
processes do not confound the
prompt-design effect.

gpt-5.4-nano is the primary agent: cheapest recent SOTA API model, fast, and in game-theoretic work smaller models tend to behave more human-like. Claude is absent only because it does not accept the same strict JSON schema through the OpenAI-compatible API.

The prompt design grid: A B C/L D

Axis	0	1	2
A — ego profile	no household info	data as a listing	LLM-written narrative
B — neighbour profile	no neighbour info	data as a listing	LLM-written narrative
C — endorsement (<i>adoption</i>)	decision not mentioned	neighbour's decision given	—
L — leader (<i>transmission</i>)	not a seed	seeded by the MFI	—
D — response format	plain	model of appropriateness	decision-theoretic

D1 — model of appropriateness. The agent answers, in order:

1. What kind of situation is this?
2. What kind of person am I?
3. What should someone like me do in a situation like this?

D2 — decision-theoretic. The agent must return a strict JSON schema populating the decision matrix of G^A/G^T — P_i over S , U_i over $A \times S$ — **with a justification per cell**, then choose.

This is what makes the reasoning auditable: the decision can be checked against the agent's *own* matrix.

Adoption: $3 \times 3 \times 2 \times 3 = 54$ designs. Transmission: the same grid with C replaced by the conditional L.

Three experiments, increasing autonomy

Pilot

Which prompt design is worth paying for?

- No simulation loop — two hand-built contrastive samples from village 6: one real adopter, one real non-adopter.
- Full factorial, both games, $n = 25$ reps/arm.
- Fisher's exact per design, FDR-corrected.
- Plus: do D2 designs **obey their own matrix**?

Hybrid

Does an LLM beat a fitted logit at take-up?

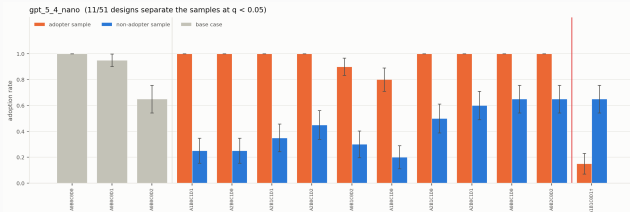
- BCDJ's Bernoulli transmission at q_N, q_P , transliterated unchanged.
- **Only adoption is an LLM call** — that single substitution is the whole experiment, which is what licenses attributing any gap to it.
- Mirror arm (LLM transmission, algorithmic adoption): NotImplementedError.

Full LLM

Can an ABM with zero fitted parameters reach a 2-parameter structural model?

- Adoption **and** per-edge transmission are both LLM calls.
- Seeded only at the empirical leaders; BCDJ's timing kept.
- Scored against real take-up *and* BCDJ's logit on the same network.
- Six models, two villages.

Pilot I — adoption



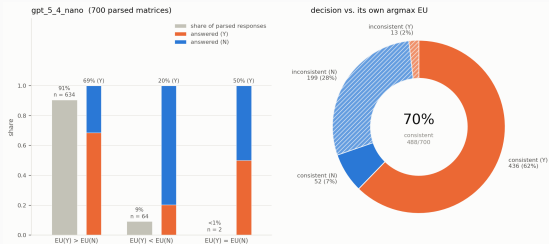
11 of 51 designs separate a real adopter from a real non-adopter at $q < 0.05$; **one inverts** (right of the red line).

Only **two** survivors are non-endorsed — the agents pick up far more from the *neighbour's decision* than from the demographic data.^a

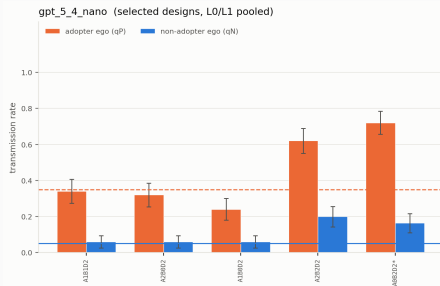
The D2 audit. 700 parsed matrices; 91% put $EU(Y) > EU(N)$, yet the decision matches the agent's own argmax only **70%** of the time — and the 28% that break it break it by answering *no* to their own yes.

Forcing reasoning through a semi-formal bottleneck buys legibility, and immediately exposes the inconsistency it made visible.

^aOn this sample pair. Needs an in-depth analysis before it becomes a claim.



Pilot II — transmission

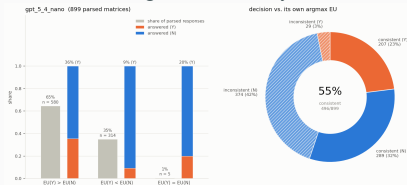


Top 5 lowest-rate designs, L0/L1 pooled. Dashed/solid lines are the paper's fitted $q_P = 0.35$ and $q_N = 0.05$.

No ground truth here — **transmission is observed nowhere in the bundle**. BCDJ never observed it either: q_N, q_P are identified from the adoption pattern with informed status integrated out. So the comparison is to the paper's fitted rates, and the test is **face validity only**.

Lean designs land close: A1B0D2 gives 0.24/0.06 against the paper's 0.35/0.05, preserving the taker/non-taker asymmetry. Richer profiles (A2B2D2, A0B2D2) inflate **both** rates and wash the asymmetry out.

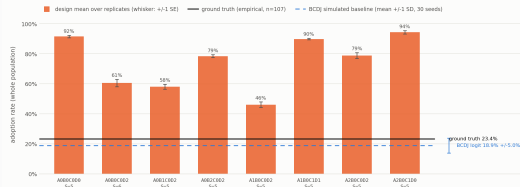
The D2 audit is worse here: 899 parsed matrices, only **55%** consistent with their own argmax — 42% say *no* where their own matrix says *yes*.^a



^a A gap our logical-enhancement frameworks are built to close.

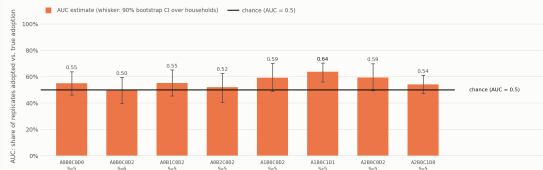
Hybrid — incomplete, and instructive

Village 6 -- adoption rate by design, against the take-up it is aiming at



denominator: full giant component village, n=101 (a household never asked was never informed and counts as not adopted; every design overshoots the village's real take-up by 2.0 to 4.0x. The baseline's skewed to +/- 1.5D over its seeds, not the SE of its mean: it stands in for one realisation of village 6, which is what a design is compared against).

Village 6 -- does a design rank households correctly, even at the wrong level?



1 of 8 designs keep their whole interval above chance (those estimates are printed in black); the CI is 500 bootstrap resamples over households, not over replicates: it says how far the AUC estimate would move under a different draw of village 6's households, not how many more replicates the design needs: a design at 5 replicates can only score a household at multiples of 1/5, so a slight 5 consensus the ranking it is being credited with.

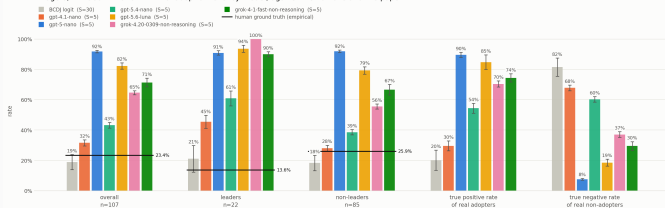
The failure, plainly. All 8 designs overshoot village 6's real take-up by $2.0\text{--}4.0\times$ ($23.4\% \rightarrow 46\text{--}94\%$); BCDJ's logit sits at 18.9%. Nor is it a ranking problem behind a level problem: **AUC** is $0.50\text{--}0.64$, only 1 of 8 keeps its interval above chance, and **8 of 8** put leaders above non-leaders where the truth is 13.6% against 25.9%.

Where the divergence may come from

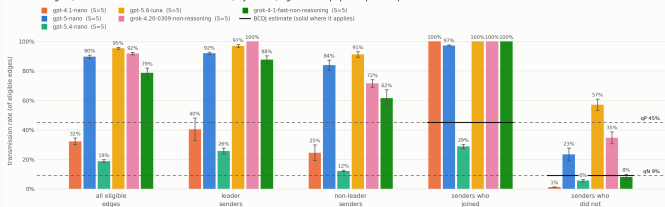
- The pilot scored a *contrast* between two curated samples; the loop demands an absolute *level*, and nothing calibrates it.
- The pilot separated mostly through endorsement (C); under BCDJ's Bernoulli flood almost everyone is informed, so C fires everywhere and stops discriminating.
- Mechanism transfer is not free: context-sampling-to-LLM and LLM-to-LLM behave differently from **algorithmic-transmission-to-LLM**.
- The mirror arm was never run, so the ablation that would localise this is missing.

Full LLM — village 6 (A1B0C0D2-A0B0D2)

Village 6, A1B0C0D2-A0B0D2 -- final adoption level by model, against the humans and the paper



Village 6, A1B0C0D2-A0B0D2 -- who tells whom, by model, against the paper's qN and qP



One design pair, six models, $n = 107$ households.

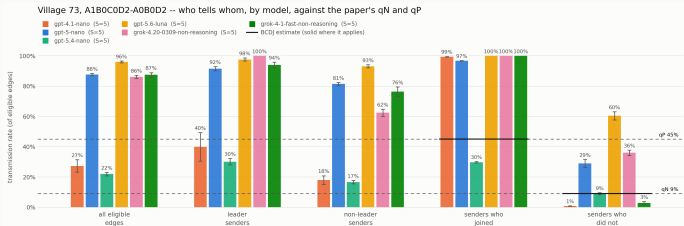
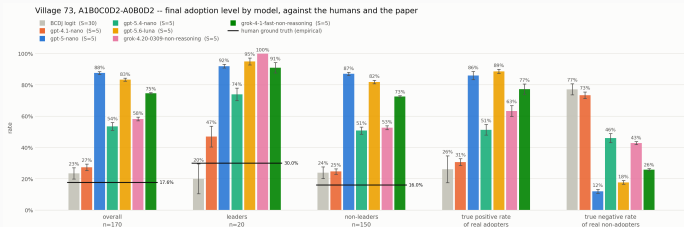
Smaller is closer. gpt-4.1-nano lands at 32% against a 23.4% truth and beats the BCDJ logit on true-positive rate. Newer gpt-5-nano / gpt-5.6-luna hit 82–92%, and their true negative rate collapses to 8–19%.

Everyone gets leaders backwards.

Truth: leaders 13.6%, non-leaders 25.9%. Every model — and BCDJ's own logit — puts leaders on top.

Transmission. gpt-5.4-nano is the only model near the paper's rates (19% vs q_P 45% / q_N 9%) and the only one keeping the taker asymmetry. The rest fire on 79–95% of eligible edges: everyone tells everyone, and the network stops mattering.

Full LLM — village 73, out of sample



Same configuration, **denser network**, $n = 170$. Nothing was re-tuned.

Why this village. Village 73 **flips the leader gap**: leaders adopt at 30.0% against non-leaders' 16.0%, the opposite of the average case. A second axis for asking whether responses are grounded in the data at all.

The honest reading. Every model puts leaders above non-leaders here too — so they now agree with the truth, but for the *same* reason they contradicted it in village 6. This is a constant prior on status, not sensitivity to the village.

The level story transfers unchanged: gpt-4.1-nano at 27% against a 17.6% truth, everything newer at 54–88%.

What could an LLM actually buy us here?

1. Mechanism, not just fit

A calibrated, verified agent produces a reasoning trace that is a *plausible* account of why this household joined — the kind of explanation a human would give in the same circumstances. Two fitted numbers, q_N and q_P , cannot say anything about why.

2. Latent heterogeneity the covariates do not carry

The best single covariate explains $R^2 = 0.009$ of adoption. If social context, in/out-group standing and who-told-you matter, an LLM is one of the few ways to enter them into the model at all — and it can generate counterfactual populations, not merely re-fit the observed one.

3. Portability without a fitted parameter

The full-LLM model has **zero fitted parameters**: it ran on village 73 untouched. A structural model must be re-estimated per setting — that is the trade this study is pricing.

Not yet demonstrated. Current results show sensitivity to social context but **no advantage over the simpler model**: levels overshoot $2-4\times$, household ranking is near chance, the leader effect is a prior rather than a reading of the data, and D2 traces contradict their own matrices 30–45% of the time. Stronger claims need reasoning ground truth.

Stretch target: a Western comparison population

The structure this study needs is **demographics + network + observed adoption**. For a WEIRD population that combination is essentially unavailable off the shelf — but experimental economics has close cousins, where the demographics are collected and “adoption” is a recorded choice under a known incentive structure.

Candidate paradigms

- **Public goods games** — contribution and conditional cooperation; the nearest analogue to take-up under social influence.
- **Sequential / multi-stage games** — trust, centipede, bargaining: the same decision-under-a-signal shape as G^A .
- **Market entry games** — coordination with a congestion externality, close to G^T .
- **Signalling games** — for the endorsement channel (C) that drove the pilot separation.

What transfers, what does not

- **Transfers**: the agent architecture, the prompt-design grid, the D2 self-consistency audit, the whole scoring apparatus.
- **Does not**: the real social network. Lab populations are anonymous and randomly rematched — the topology would have to be commissioned, or recovered from a field experiment.

Centaur (Binz et al., 2024) is the proof the direction is tractable: one model fine-tuned across 160 experiments, predicting held-out human behaviour including in *unseen* paradigms. The natural benchmark — and a natural source of the behavioural ground truth this study lacks.

Thank you